(iv)
$$\left.\begin{aligned} x+\ y+\ z+\ t&=5\\ 2x+\ y+3z-\ t&=14\\ 3x+3y-2z+2t&=1\\ 4x-2y+\ z-3t&=6 \end{aligned}\right\}$$

Answer : $x=1, y=2, z=3, t=-1.$

38. Find the inverse of $A=\begin{bmatrix}1 & -1 & 2 & 1\\ 3 & 0 & 2 & 2\\ 2 & 1 & -1 & 1\\ 1 & 0 & 1 & 1\end{bmatrix}$ [C. U. P. 1978]

Answer : $A^{-1}=\begin{bmatrix}0 & 1 & 0 & -2\\ -2 & 1 & -1 & 1\\ -1 & 1 & -1 & 0\\ 1 & -2 & 1 & 3\end{bmatrix}$

39. Find the inverse of the matrix $A=\begin{bmatrix}1 & 2 & 1 & 2 & 3\\ 2 & 3 & 1 & 0 & 1\\ 2 & 2 & 1 & 0 & 0\\ 1 & 1 & 1 & 1 & 1\\ 0 & -2 & 0 & 2 & -2\end{bmatrix}$

by using only the elementary row operations (row equivalent canonical matrix).

Answer :

$$A^{-1}=\begin{bmatrix}2 & -4 & 5 & -3 & -\frac{1}{2}\\ -1 & 3 & -3 & 1 & \frac{1}{2}\\ -2 & 2 & -3 & 4 & 0\\ 0 & 1 & -1 & 0 & \frac{1}{2}\\ 1 & -2 & 2 & -1 & \frac{1}{2}\end{bmatrix}$$

40. If A and B are two symmetric matrices of the same order, then show that a necessary and sufficient condition for the matrix AB to be symmetric is that AB = BA.

41. Show that every non-singular idempotent matrix is an identity matrix.

42. If A is an idempotent matrix and A + B = I, then prove that B is an idempotent matrix and AB = BA = 0.

43. If A and B are n-rowed unitary matrices, then prove that BA is also unitary matrix.

(iii) $\left.\begin{aligned} 3x+2y-z&=20\\ 2x+3y-3z&=7\\ x-y+6z&=41 \end{aligned}\right\}$ [C. U. P. 1977

(iv) $\left.\begin{aligned} 3x-y+z&=-2\\ x+5y+2z&=6\\ 2x+3y+z&=0 \end{aligned}\right\}$ (v) $\left.\begin{aligned} x+2y+z&=2\\ 2x-y+2z&=-1\\ 3x-4y-3z&=-16 \end{aligned}\right\}$

Answers : (i) $x=2; y=3,$ (ii) $x=1=y=2, z=3.$

(iii) $x=5, y=6, z=7,$ (iv) $x=-2, y=0, z=4,$

(v) $x=-2, y=1, z=2.$

36. Solve the following systems of linear equations with the help of matrices :

(i) $\left.\begin{aligned} 5x-6y+4z&=15\\ 7x+4y-3z&=19\\ 2x+y+6z&=46 \end{aligned}\right\}$ [D. U. P. 1979, J. U. H. 1980]

Answer : $x=3, y=4, z=6.$

(ii) $\left.\begin{aligned} 2x-3y+4z&=1\\ 3x+4y-5z&=10\\ 5x-7y+2z&=3 \end{aligned}\right\}$ [D. U. P. 1966. J. U. H. 1978]

Answer : $x=2, y=1, z=0.$

(iii) $\left.\begin{aligned} x+y+z&=9\\ 2x+5y+7z&=52\\ 2x+y-z&=0 \end{aligned}\right\}$ [C. U. P. 1984]

Answer : $x=1, y=3, z=5.$

(iv) $\left.\begin{aligned} x+2y-z&=-1\\ 3x+8y+2z&=28\\ 4x+9y-z&=14 \end{aligned}\right\}$ [D. U. H. 1976]

Answer : $x=-26, y=13, z=1.$

(v) $\left.\begin{aligned} x_1+3x_2+2x_3&=5\\ 2x_1+x_2+3x_3&=1\\ 3x_1+2x_2+x_3&=4 \end{aligned}\right\}$ [D. U. H. 1978]

Answer : $x_1=\frac{2}{9}, x_2=\frac{17}{9}, x_3=-\frac{4}{9}.$

(vi) $\left.\begin{matrix} x+y+z=1 \\ 2x+5y+5z=2 \\ 4x+9y+12z=3 \end{matrix}\right\}$ [D. U. P. 1975]

Answer : $x=1, y=\frac{1}{3}, z=-\frac{1}{3}$.

(vii) $\left.\begin{matrix} 2x-y-z=6 \\ x+3y+2z=1 \\ 3x-y-5z=1 \end{matrix}\right\}$ [D. U. P. 1985]

Answer : $x=3, y=-2, z=2$.

(viii) $\left.\begin{matrix} x+y+z=3 \\ x+2y+2z=4 \\ x+4y+9z=6 \end{matrix}\right\}$ [D. U. H. 1983]

Answer : $x=2, y=1, z=0$.

(ix) $\left.\begin{matrix} 2x+y+z=3 \\ 3x-2y+2z=9 \\ 4x+3y-z=-1 \end{matrix}\right\}$

Answer : $x=1, y=-1, z=2$.

(x) $\left.\begin{matrix} x+2y+3z+4=0 \\ 2x+4y+5z+7=0 \\ 3x+5y+6z+10=0 \end{matrix}\right\}$ [D. U. P. 1986]

Answer : $x=-3, y=1, z=-1$.

37. Solve the following systems of linear equations by using row equivalent canonical matrix (by elementary row transformations) :

(i) $\left.\begin{matrix} 2x-y+z=1 \\ x+4y-3z=-2 \\ 3x+2y-z=0 \end{matrix}\right\}$ (ii) $\left.\begin{matrix} x+2y+z=2 \\ 3x+y-2z=1 \\ 4x-3y-z=3 \\ 2x+4y+2z=4 \end{matrix}\right\}$

(iii) $\left.\begin{matrix} x+2y+3z+t=3 \\ x+y+z-t=5 \\ x+y-z+t=-4 \\ x-y+z+t=2 \end{matrix}\right\}$

Answers : (i) $x=0, y=1, z=2$. (ii) $x=1, y=0, z=1$.

(iii) $x=1, y=-1, z=2, t=-2$.

(v) $\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$

27. If $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$ then show that AA^T and A^TA are both symmetric matrices.

28. Show that the matrix

$$A = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$ is orthogonal.

29. Prove that the matrix

$$A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$ is unitary.

30. Show that the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is idempotent.

31. Show that the matrix $A = \begin{bmatrix} 4 & -1 & -4 \\ 3 & 0 & -4 \\ 3 & -1 & -3 \end{bmatrix}$ is involutory.

32. Show that if $A = \begin{bmatrix} 0 & 4 & 3 \\ 1 & -3 & -3 \\ -1 & 4 & 4 \end{bmatrix}$ then $A^2 = I$. → involutory

33. Show that if $A = \begin{bmatrix} -1 & 1 & -1 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{bmatrix}$ then $A^2 = A$. → idempotent

34. Prove that if $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ then $A^4 = 0$, but $A^3 \neq 0$.

35. Solve the following systems of linear equations with the help of matrices :

(i) $\left.\begin{aligned} x + y &= 5 \\ 2x + 3y &= 13 \end{aligned}\right\}$ (ii) $\left.\begin{aligned} x + y + z &= 6 \\ x - y + z &= 2 \\ 2x + y - z &= 1 \end{aligned}\right\}$ [D. U. P. 1977

20. Find the inverse of each of the following matrices by using only the elementary row operations (row equivalent canonical matrix) :

(i) $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 3 & -4 \\ 1 & 2 & 3 \\ 3 & -1 & -1 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 3 & 4 \\ 3 & -1 & 6 \\ -1 & 5 & 1 \end{bmatrix}$

(iv) $\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$ (v) $\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 0 & 2 & 2 \\ 2 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ (vi) $\begin{bmatrix} 2 & 5 & 2 & 3 \\ 2 & 3 & 3 & 4 \\ 3 & 6 & 3 & 2 \\ 4 & 12 & 0 & 8 \end{bmatrix}$

Answers :

(i) $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 7 & 17 \\ 10 & 10 & -10 \\ -7 & 11 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} \frac{31}{2} & -\frac{17}{2} & -11 \\ \frac{9}{2} & -\frac{5}{2} & -3 \\ -7 & 4 & 5 \end{bmatrix}$

(iv) $\begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}$ (v) $\begin{bmatrix} 0 & 1 & 0 & -2 \\ -2 & 1 & -1 & 1 \\ -1 & 1 & -1 & 0 \\ 1 & -2 & 1 & 3 \end{bmatrix}$

(vi) $\begin{bmatrix} -3 & \frac{3}{4} & \frac{5}{4} & \frac{7}{16} \\ 1 & -\frac{5}{12} & -\frac{1}{4} & \frac{5}{48} \\ 1 & -\frac{1}{12} & -\frac{1}{4} & -\frac{13}{48} \\ 0 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{16} \end{bmatrix}$

21. If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & -1 & 1 \end{bmatrix}$ then find $A^3 - 2A^2 - 4A - 2I$ and hence find A^{-1}.

[D. U. H. T. 1984]

Answers : Θ and $A^{-1} = \frac{1}{2}\begin{bmatrix} 3 & -5 & 9 \\ 1 & -1 & 1 \\ -2 & 4 & -6 \end{bmatrix}$.

22. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

Answer : $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$.

23. Show that the matrix

$$A = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is invertible for all values of θ and find A^{-1}.

Answer : $A^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

24. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$

are two matrices, then find $\bar{A}^1 B$ and $A \bar{B}^1$.

[D. U. H. T. 1981, 1987]

Answers : $\bar{A}^1 B = \frac{1}{3}\begin{bmatrix} 3 & -3 & -11 \\ 0 & 3 & 10 \\ 0 & 0 & -2 \end{bmatrix}$, $A \bar{B}^1 = \frac{1}{2}\begin{bmatrix} 0 & -2 & 0 \\ -2 & 4 & 0 \\ -12 & 17 & -3 \end{bmatrix}$

25. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

then prove that $(AB)^{-1} = \bar{B}^1 . \bar{A}^1$ **[D. U. H. T. 1983, 1987]**

26. Prove that the following matrices are orthogonal :

(i) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ (ii) $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$ (iv) $\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

15. If $A=\begin{bmatrix}2 & 3 & 4\\ 4 & 3 & 2\\ 2 & 4 & 3\end{bmatrix}$ and $B=\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\\ 3 & 4 & 1\end{bmatrix}$

Show that $(A+B)^2 \neq A^2+2AB+B^2$.

[D. U. H. 1976, C. U. H. 1977]

16. If $A=\begin{bmatrix}1 & 2 & 3\\ 1 & 3 & 5\\ 1 & 5 & 12\end{bmatrix}$ and $B=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 1 & 4 & 9\end{bmatrix}$

Find the values of $A^{-1}B$ and AB^{-1} **[D. U. P. 1976]**

Answers : $A^{-1}B=\begin{bmatrix}1 & -1 & -\frac{7}{3}\\ 0 & 1 & \frac{2}{3}\\ 0 & 0 & \frac{2}{3}\end{bmatrix}$ $AB^{-1}=\begin{bmatrix}0 & 1 & 0\\ 1 & 2 & 0\\ 0 & -\frac{1}{2} & \frac{3}{2}\end{bmatrix}$

17. If $A=\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$ Find $\frac{1}{2}(A^{-1}+A)$. **[D. U. P. 1977]**

Answer : $\frac{1}{2}(A^{-1}+A)=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=I.$

18. If $A=\begin{bmatrix}3 & -3 & 4\\ 2 & -3 & 4\\ 0 & -1 & 1\end{bmatrix}$ Prove that $A^3=A^{-1}$

19. Find the inverse of the matrix

$$A=\begin{bmatrix}1 & 0 & 1 & 2\\ -1 & 1 & 2 & 1\\ 1 & 0 & 1 & 1\\ 3 & 1 & 0 & 2\end{bmatrix}.$$

Answer : $A^{-1}=\begin{bmatrix}-\frac{1}{2} & -\frac{1}{6} & \frac{5}{6} & \frac{1}{6}\\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{1}{6} & \frac{7}{6} & -\frac{1}{6}\\ 1 & 0 & -1 & 0\end{bmatrix}$

12. (i) Find the inverse of the matrix

$$A=\begin{bmatrix}2 & -1 & -1\\ 1 & -2 & 1\\ 1 & -1 & 2\end{bmatrix}$$ [J. U. H. 1975]

(ii) Find the inverse of the matrix

$$A=\begin{bmatrix}5 & -6 & 4\\ 7 & 4 & -3\\ 2 & 1 & 6\end{bmatrix}$$ [D. U. P. 1979]

Answers : (i) $A^{-1}=\begin{bmatrix}\frac{1}{2} & -\frac{1}{2} & \frac{1}{2}\\ \frac{1}{6} & -\frac{5}{6} & \frac{1}{2}\\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2}\end{bmatrix}$

(ii) $A^{-1}=\frac{1}{419}\begin{bmatrix}27 & 40 & 2\\ -48 & 22 & 43\\ -1 & -17 & 62\end{bmatrix}$

13. If $A=\begin{bmatrix}1 & -3 & 2\\ 4 & 1 & -1\\ -3 & 2 & 5\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1 & 5\\ -1 & -2 & -2\\ 3 & 1 & 2\end{bmatrix}$

Find the matrices AB, BA, A^{-1}, B^{-1}, $(AB)^{-1}$. Check your results by verifying the relations.

$(AB)^{-1}=B^{-1}A^{-1}$, $AA^{-1}=I$, $B^{-1}B=I$.

Answers : $AB=\begin{bmatrix}11 & 9 & 15\\ 4 & 1 & 16\\ 7 & -2 & -9\end{bmatrix}$, $BA=\begin{bmatrix}-9 & 5 & 28\\ -3 & -3 & -10\\ 1 & -4 & 15\end{bmatrix}$,

$$A^{-1}=\frac{1}{80}\begin{bmatrix}7 & 19 & 1\\ -17 & 11 & 9\\ 11 & 7 & 13\end{bmatrix},$$

$$B^{-1}=\frac{1}{17}\begin{bmatrix}-2 & 3 & 8\\ -4 & -11 & -1\\ 5 & 1 & -3\end{bmatrix}$$

$$(AB)^{-1}=\frac{1}{1360}\begin{bmatrix}23 & 51 & 129\\ 148 & -204 & -116\\ -15 & 85 & -25\end{bmatrix}$$

14. If $A=\begin{bmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}0 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$

Show that $A^3-A^2+A-2B=0$. [C. U. P. 1976]